The Cato Institute's Overlapping Generations Model for Social Security Reform

Krit Chanwong

March 2026

1 Introduction

The aim of this model is to provide the public with a computable overlapping generations (OLG) model that is capable of scoring most Social Security reforms. The OLG model we have built here takes most of its structure from the Penn-Wharton Budget Model's (PWBM) OLG. However, we make multiple important changes to the PWBM's OLG, as will be noted.

2 Production/Government

2.1 Technology

Our production function is a simple Cobb-Douglas with no technology:

$$F(K, L) = K^{\alpha} L^{1-\alpha} = Y$$

Where K is aggregate capital, L is aggregate labor in efficiency units, and $\alpha = 0.36$ is the capital share. Firms are competitive and hire factors up to their marginal products:

$$w = \frac{\partial F}{\partial L} = (1-\alpha)\left(\frac{K}{L}\right)^{\alpha}, \qquad r_K = \frac{\partial F}{\partial K} = \alpha\left(\frac{L}{K}\right)^{1-\alpha}$$

2.2 Breakdown of Output

We can break up output into four distinct parts:

$$Y = y_{lab,k} + y_{corp} + y_{pass} + y_{debt}$$

In this formulation, $y_{lab,k}$ represents labor income, which is as follows:

$$y_{\text{lab},k} = w \cdot z_k \cdot n_k$$

Households hold a single asset a. The composition of the economy's asset pool determines how each household's savings map into different income streams. Define the portfolio shares:

$$\psi_{\text{cap}} = \frac{K}{K+D}, \qquad \psi_{\text{debt}} = \frac{D}{K+D}$$

where D is government debt. A fraction ψ_{cap} of each household's assets is implicitly invested in productive capital, and ψ_{debt} in government bonds. There is no portfolio choice—all households hold the same mix.

The capital share is further split between the corporate sector (fraction $\zeta_{\text{corp}} = 0.55$) and the pass-through sector (fraction $\zeta_{\text{pass}} = 0.45$). Both sectors earn the net-of-depreciation return $\pi = r_K - \delta$, but face different tax treatment. This yields three capital income flows:

$$y_{\text{corp}} = \pi \cdot \zeta_{\text{corp}} \cdot \psi_{\text{cap}} \cdot a \tag{1}$$

$$y_{\text{pass}} = \pi \cdot \zeta_{\text{pass}} \cdot \psi_{\text{cap}} \cdot a \tag{2}$$

$$y_{\text{debt}} = r_G \cdot \psi_{\text{debt}} \cdot a \tag{3}$$

where $r_G = 0.03$ is the exogenous government bond rate and $\delta = 0.06$ is the depreciation rate.

3 Taxation

The tax function τ in this model mirrors the structure of the U.S. federal tax code. Total taxes paid by a household consist of four components:

$$\tau = T_{\text{ord}} + T_{\text{pref}} + T_{\text{payroll}} + T_{\text{con}}$$

3.1 Income Classification

Each income source is assigned an *inclusion rate* that determines what fraction enters the taxable base. These inclusion rates approximate the effect of deductions and exclusions without explicitly modeling itemization:

Income Source	Ordinary Inclusion (θ^{ORD})	Preferential Inclusion (θ^{PREF})
Labor income	0.95	—
Corporate equity	0.60	0.40
Pass-through	1.00	—
Bond interest	1.00	—
Social Security	Formula	—

The 95% inclusion rate on labor income approximates the standard deduction. Corporate equity income is split between ordinary income (dividends taxed at ordinary rates) and preferential income (long-term capital gains and qualified dividends).

3.2 Social Security Benefit Taxation

The taxable portion of Social Security benefits is determined by the modified adjusted gross income (M-AGI) test, which is:

$$M_{AGI} = y_{\text{other}} + 0.5 \cdot b^{SS}$$

where y_{other} is all non-Social-Security ordinary income and b^{SS} is total Social Security benefits. The taxable amount is then:

$$b^{SS}_{\text{taxable}} = \begin{cases} 0 & \text{if } M_{AGI} \leq \$32{,}000 \\ \min\left(0.50 \cdot b^{SS},\ 0.50 \cdot (M_{AGI} - \$32{,}000)\right) & \text{if } \$32{,}000 < M_{AGI} \leq \$44{,}000 \\ \min\left(b^{SS}_{50\%} + 0.85 \cdot (M_{AGI} - \$44{,}000),\ 0.85 \cdot b^{SS}\right) & \text{if } M_{AGI} > \$44{,}000 \end{cases}$$

where $b^{SS}_{50\%}$ denotes the tier-1 taxable amount. The thresholds \$32,000 and \$44,000 are the married-filing-jointly thresholds. Critically, these thresholds are *not indexed* to inflation or wage growth

The taxable amount is further decomposed into an OASI component (the first 50% tier) and an HI component (the remainder up to the 85% tier), since the resulting tax revenues are credited to different trust funds.

3.3 Ordinary Income Tax

Total ordinary income is:

$$y_{\text{ord}} = \theta^{ORD}_{\text{lab}} \cdot y_{\text{lab}} + \theta^{ORD}_{\text{corp}} \cdot y_{\text{corp}} + \theta^{ORD}_{\text{pass}} \cdot y_{\text{pass}} + y_{\text{debt}} + b^{SS}_{\text{taxable}}$$

This is taxed through a progressive bracket schedule. For married-filing-jointly households:

Bracket Floor	Marginal Rate
\$0	10%
\$19,400	12%
\$78,950	22%
\$168,400	24%
\$321,450	32%
\$408,200	35%
\$612,350	37%

Survivors (widows/widowers) file as single filers and face narrower brackets with lower thresholds. The ordinary income tax is computed as:

$$T_{\text{ord}} = \sum_i r_i \cdot \max\left(0,\ \min(y_{\text{ord}},\ b_{i+1}) - b_i\right)$$

where r_i is the marginal rate and b_i is the floor of bracket i.

3.4 Preferential Income Tax

The preferential portion of corporate equity income (40% of corporate income, capturing long-term capital gains and qualified dividends) is taxed at lower rates:

Bracket Floor (MFJ)	Rate
\$0	0%
\$78,750	15%
\$488,850	20%

3.5 Payroll Tax

Each spouse's labor income is subject to a payroll tax up to the taxable maximum:

$$T_{\text{payroll}} = \tau_p \cdot \min(y_{\text{lab},1},\, \overline{y}) + \tau_p \cdot \min(y_{\text{lab},2},\, \overline{y})$$

where $\tau_p = 0.106$ is the combined employer-employee OASI rate and $\overline{y} =$ \$176,100 is the taxable earnings cap. The cap is indexed to average wage growth in the projection. Note that this rate covers only the OASI portion of FICA; the HI (Medicare) component is handled separately through the benefit taxation channel.

3.6 Marginal Tax Rates

For the household's labor supply decision, the relevant object is the marginal tax rate on labor income. Because the progressive bracket structure, the payroll tax cap, and the Social Security benefit taxation formula all interact, the marginal tax rate is computed numerically:

$$MTR_k = \frac{\tau(y_{\text{lab},k}, \ldots) - \tau(y_{\text{lab},k}, \ldots)}{\epsilon} + \mathbf{1}(y_{\text{lab},k} < \overline{y}) \cdot \tau_p$$

This captures the fact that an additional dollar of labor income can simultaneously push the household into a higher bracket *and* cause a larger share of Social Security benefits to become taxable, generating effective marginal rates that exceed the statutory bracket rate.

3.7 Corporate Income Tax

On the production side, corporate profits are taxed at a statutory rate of $\tau_c = 0.21$. The effective corporate tax base accounts for partial expensing of investment ($\phi_{\text{exp}} = 0.50$), full interest deductibility ($\phi_{\text{int}} = 1.00$), and other deductions and credits. This tax does not enter the household optimization directly but affects the general equilibrium through the cost of capital and the government budget constraint.

3.8 Government Sector

The government here receives money from income, payroll, and corporate taxes. It uses this money on Social Security and other undetermined government spending. Like the closed version of the Penn-Wharton Budget Model, the government here balances its budget through raising extra revenues, the source of which is undetermined.

4 Social Security

4.1 Average Indexed Monthly Earnings

Social Security benefits are computed from each worker's Average Indexed Monthly Earnings (AIME). For each spouse, the model records annual earnings $e_{k,j} = w \cdot z_k \cdot n_k$ over the working life, subject to the taxable maximum. Earnings prior to age 60 are indexed to economy-wide wage growth:

$$\tilde{e}_{k,j} = \begin{cases} \min\left(e_{k,j} \cdot (1+g)^{60-j},\ \overline{y}\right) & \text{if } j < 60 \\ \min\left(e_{k,j},\ \overline{y}\right) & \text{if } j \geq 60 \end{cases}$$

The AIME is then the average of the highest 35 years of indexed earnings:

$$\text{AIME}_k = \frac{1}{35} \sum_{i=1}^{35} \tilde{e}_{k,(i)}$$

where $\tilde{e}_{k,(i)}$ denotes the i-th highest indexed earning year.

4.2 Primary Insurance Amount

The Primary Insurance Amount (PIA) is computed from AIME using the standard bend-point formula:

$$\text{PIA} = 0.90 \cdot \min(\text{AIME}, b_1) + 0.32 \cdot \max(0, \min(\text{AIME}, b_2) - b_1) + 0.15 \cdot \max(0, \text{AIME} - b_2)$$

where $b_1 = \$14{,}712$ and $b_2 = \$88{,}692$ are the bend points (annualized) in 2025.

4.3 Trust Fund Accounting

In the projection engine, the OASI trust fund evolves as:

$$TF_{t+1} = TF_t + \underbrace{\tau_p \cdot Y_t^{\text{taxable}}}_{\text{FICA revenue}} + \underbrace{T_{\text{btax},t}^{OASI}}_{\text{benefit tax revenue}} + \underbrace{r_{TF} \cdot TF_t}_{\text{interest}} - \underbrace{S_t}_{\text{outlays}}$$

where $TF_0 = \$2.76$ trillion, $r_{TF} = 4.7\%$, and interest accrues only while the trust fund is positive. The trust fund depletion year is the first year in which $TF_t < 0$.

5 Household Problem and Solution

Each person has the following flow utility:

$$u(c,n) = \frac{c^{1-\sigma}-1}{1-\sigma} + \gamma \ln(1-n)$$

where c denotes consumption, $n \in [0,1)$ denotes hours worked as a fraction of the time endowment, σ is the coefficient of relative risk aversion, and γ governs the taste for leisure. We set $\sigma = 2$ and $\gamma = 0.30$.

The representative household in this model consists of a married couple. Both spouses consume a single household good c and independently supply labor. The couple maximizes a unitary household utility function:

$$U(c,n_1,n_2) = \frac{c^{1-\sigma}-1}{1-\sigma} + \gamma\left[\ln(1-n_1) + \ln(1-n_2)\right]$$

where n_1 and n_2 are the hours worked by spouse 1 and spouse 2, respectively.

Personal income is the product of three components: productivity z, wages w, and hours worked n. In this model, individual productivity is the sum of the age-based efficiency component $z_{a,j}$, a permanent income process $z_{i,.}$, and a persistent process (z_{ij}). In other words:

$$\ln(z) = z_{a,j} + z_{i,.} + z_{ij}$$

The age-efficiency parameter, $z_{a,j}$, is determined by a quadratic production efficiency function. The persistent process $z_{i,j}$ is determined by an AR(1) process, such that $z_{i,j} = \rho z_{i-1,j} + \eta$, where $\eta \sim N(0,\sigma_\eta^2)$. Like the Penn-Wharton model, we set the ρ parameter to 0.990. For computational ease, we discretize this AR(1) process with Tauchen's method. The permanent income process $z_{i,.}$, is determined by a probability mass function, which is randomly determined. This PMF determines lifetime productivity with respect to average earnings.

Once productivity has been determined for a household, wages can be ascertained through using the marginal product of the production function. The only part remaining is labor supply, which is determined by the intratemporal first-order condition:

$$n_{ij} = 1 - \frac{\gamma}{wz(1-MTR)\cdot c^{-\sigma}}$$

This expression equates the marginal disutility of labor, $\gamma/(1-n)$, to the after-tax marginal return to working, $wz(1-MTR)$, valued at the marginal utility of consumption $c^{-\sigma}$.

The representative household here has two members. A household's gross income is simply the sum of both members' labor income and the return on assets:

$$\text{Inc}_{\text{household}} = z_{ij}^1 n_{ij}^1 w + z_{ij}^2 n_{ij}^2 w + (1+\pi)a$$

We assume here that there are three types of asset classes: bonds, corporate equity, and pass-through income. Each of these has a return, which is based on

the taxation of each of these asset classes. As such, we can decompose household income as:

$$Inc_{household} = z^1_{ij} n^1_{ij} w + z^2_{ij} n^2_{ij} w +$$
$$(\pi_{corp}\zeta_{corp}\psi_{cap} + \pi_{pass}\zeta_{pass}\psi_{cap} + r_G\psi_{debt})a$$

5.1 EGM Solution

This household has the following budget constraint:

$$c_t + a_{t+1} + \tau(z^1_{ij} n^1_{ij} w, z^2_{ij} n^2_{ij}, c_t, a_t) = z^1_{ij} n^1_{ij} w_t + z^2_{ij} n^2_{ij} w_t + (1+\pi)a_t$$

Where τ is the tax function. The Bellman equation for this problem becomes:

$$V^c(a, z_1, z_2, j) = \max_{c, n_1, n_2, a'} \; U(c, n_1, n_2) + \beta \, E_j \left[V_{j+1}\right]$$

subject to the budget constraint:

$$c_t + a_{t+1} + \tau(z^1_{ij} n^1_{ij} w, z^2_{ij} n^2_{ij}, c_t, a_t) = z^1_{ij} n^1_{ij} w_t + z^2_{ij} n^2_{ij} w_t + (1+\pi)a_t$$

.

Here τ is the taxation function, which is described in Section 3. The continuation value reflects gender-specific mortality. At each age j, either both spouses survive, only the husband survives, or only the wife survives:

$$E_j \left[V_{j+1}\right] = s^1_j s^2_j \, E\left[V^c_{j+1}\right]$$
$$+ s^1_j (1 - s^2_j) \, E\left[V^{s_1}_{j+1}\right] + (1 - s^1_j) s^2_j \, E\left[V^{s_2}_{j+1}\right]$$

Survival probabilities are retrieved from the SSA's 2022 life tables. In order to solve the household problem, we use backward induction and value function iteration. Specifically, we assume that at the terminal period n_j, households consume everything. We then solve backwards for age-specific equilibrias.

We also employ endogenous grid method (EGM) in order to avoid root finding and numerical instabilities in our value-function iteration. EGM relies on an analytical expression of the Euler equation with respect to consumption. This is provided below (proof and algorithm detail in Appendix):

$$c_t^{-\sigma} = \beta(1+\pi)[s^m_j s^f_j \sum_{z'_1, z'_2} \Pi_{z_1, z'_1} \Pi_{z_2, z'_2} \left(c^1_{t+1} + c^2_{t+1}\right)^{-\sigma}$$
$$+ s^m_j (1 - s^f_j) \sum_{z'_1} \Pi_{z_1, z'_1} \left(c^{s_1}_{t+1}\right)^{-\sigma} + (1 - s^m_j) s^f_j \sum_{z'_2} \Pi_{z_2, z'_2} \left(c^{s_2}_{t+1}\right)^{-\sigma}]$$

where $c_{t+1} + c_{t+2}$, $c^{s_1}_{t+1}$, and $c^{s_2}_{t+1}$ denote next-period consumption under the couple and widowed scenarios, respectively.

6 Projection Methodology

The projection engine takes the steady-state solution and extrapolates it forward over a 75-year horizon (2025–2099), applying exogenous growth paths and demographic shifts to produce nominal fiscal aggregates.

6.1 Exogenous Growth Paths

Three exogenous growth rates drive the projection. These are derived from Social Security Trustee's Report Intermediate Scenario projections in 2026:

Variable	Symbol	Baseline Value
Productivity growth	g_A	1.2%
Population growth	g_n	0.5%
Inflation (CPI)	g_π	2.3%

From these, cumulative indices are constructed:

$$A_t = \prod_{\tau=1}^{t}(1 + g_{A,\tau}) \qquad \text{(cumulative productivity)} \tag{4}$$

$$P_t = \prod_{\tau=1}^{t}(1 + g_{n,\tau}) \qquad \text{(cumulative population)} \tag{5}$$

$$\Pi_t = \prod_{\tau=1}^{t}(1 + g_{\pi,\tau}) \qquad \text{(cumulative price level)} \tag{6}$$

$$W_t = \prod_{\tau=1}^{t}(1 + g_{A,\tau} + g_{\pi,\tau}) \qquad \text{(cumulative nominal wage)} \tag{7}$$

These growth rates can vary over the projection horizon through user-specified scenario episodes, allowing the model to accommodate recessions, productivity slowdowns, or inflationary shocks in particular year ranges.

6.2 Demographics: Dependency Ratio

The ratio of retirees to workers evolves along a logistic path:

$$d_t = d_0 + \frac{d_\infty - d_0}{1 + \exp\left(-\kappa(t - \bar{t})\right)}$$

where d_0 is the initial dependency ratio (from the steady state, approximately 0.29), $d_\infty = 0.45$ is the long-run ratio, $\kappa = 0.08$ governs the speed of transition, and $\bar{t} = 25$ is the midpoint year. This S-shaped path captures the aging of the baby boom generation and the gradual rise in the old-age dependency ratio projected by SSA and CBO.

The dependency ratio scales the number of retirees relative to the steady state:

$$N_t^{ret} = N_{ss}^{ret} \cdot \frac{d_t}{d_{ss}} \cdot N_{hh,t}$$

6.3 Revenue Projections

Payroll Tax Revenue

Total payroll grows with productivity, prices, and population. The taxable share adjusts for the fraction of earnings above the cap:

$$Y_t^{taxable} = Y_{payroll,t}^{total} \cdot (1 - p_t^{above})$$

where p_t^{above} is the fraction of total payroll above the taxable maximum, adjusted for wage growth relative to cap growth. FICA revenue is then:

$$R_t^{FICA} = \tau_p \cdot Y_t^{taxable}$$

The taxable earnings cap is indexed to nominal wage growth: $\overline{y}_t = \overline{y}_0 \cdot W_t$.

Benefit Taxation Revenue

Because the provisional income thresholds ($32,000 and $44,000) are not indexed, an increasing share of retirees' benefits becomes taxable over time. The model captures this bracket creep using a logistic approximation:

$$f_t^{OASI} = \frac{1}{1 + \left(\frac{1}{f_0^{OASI}} - 1\right) \exp\left(-\theta_s \ln \Pi_t\right)}$$

where f_0^{OASI} is the baseline fraction of benefits subject to OASI taxation and $\theta_s = 0.3$ is a sensitivity parameter. As the price level Π_t rises, the effective taxation rate on benefits converges toward the 85% statutory maximum. An analogous expression applies to the HI tier, with a lower sensitivity $(0.6 \cdot \theta_s)$ reflecting the wider gap between the second threshold and typical benefit levels.

6.4 Outlay Projections

Average Benefit Calculation

Benefits for new retirees are indexed to cumulative productivity growth (wage-indexing of the AIME and bend points). But the stock of current retirees receives cost-of-living adjustments (COLAs) tied to inflation. The model blends these through a vintage-averaging approach:

$$\bar{b}_t^{real} = (1 - \phi_t) \cdot \bar{b}_{t-1}^{real} + \phi_t \cdot b_t^{new,real}$$

where $\phi_t = \min\left(0.20, \frac{1}{\text{working years} \times d_t}\right)$ is the fraction of retirees who are new claimants in year t, and $b_t^{new,real}$ is the real benefit level for new retirees (growing with productivity). The nominal average benefit is:

$$\bar{b}_t^{nom} = \bar{b}_t^{real} \cdot \mu \cdot \Pi_{t-1}$$

using the lagged price level to reflect the one-year COLA delay. Total outlays are:

$$S_t = \bar{b}_t^{nom} \cdot N_t^{ret}$$

6.5 Trust Fund Dynamics

The OASI trust fund evolves as described in Section 4.3. Cash flow (excluding interest) is:

$$CF_t = R_t^{FICA} + R_{btax,t}^{OASI} - S_t$$

Total income includes interest on the trust fund balance (earned only while positive):

$$I_t^{TF} = \begin{cases} r_{TF} \cdot TF_t & \text{if } TF_t > 0 \\ 0 & \text{otherwise} \end{cases}$$

The trust fund updates as:

$$TF_{t+1} = TF_t + CF_t + I_t^{TF}$$

The *depletion year* is the first t such that $TF_t < 0$. The *cash-flow deficit year* is the first t such that $CF_t < 0$.

6.6 Reform Scoring

To score a reform, the model solves two separate steady states: one under current law and one under the reform. The projection then blends between the two using a linear phase-in parameter λ_t:

$$\lambda_t = \begin{cases} 0 & \text{if } t < t^{reform} \\ \min\left(1,\ \frac{t - t^{reform}}{T^{phase}}\right) & \text{if } t \geq t^{reform} \end{cases}$$

where $T^{phase} = 10$ years is the default phase-in period. Every fiscal variable x_t is computed as:

$$x_t = (1 - \lambda_t) \cdot x_t^{base} + \lambda_t \cdot x_t^{reform}$$

This applies to the payroll tax rate, the fraction of earnings above the cap, income and corporate tax revenue, benefit levels, and benefit taxation rates. The phase-in captures the fact that Social Security reforms typically grandfather existing beneficiaries and affect only new claimants gradually.

6.7 Output Metrics

The projection produces the following year-by-year series: nominal GDP, average earnings, the payroll tax cap, taxable payroll, FICA revenue, SS outlays, average benefit per retiree, benefit tax revenue (OASI and HI), trust fund interest, cash flow, total trust fund balance, effective FICA rate, cost rate (outlays as a percent of taxable payroll), debt, and debt-to-GDP. Summary statistics are reported at 10-year, 30-year, and 75-year actuarial windows.